Hints: Series and infinite products – 1. Convergence: the standard scale and comparison

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Raabe: compare with n^{-q}, whose ratio is $\left(\frac{n}{n+1}\right)^q = 1 - \frac{q}{n} + O(n^{-2})$. If $p > 1$, take $1 < q < p$: eventually $\frac{a_{n+1}}{a_n} \leqslant \left(\frac{n}{n+1}\right)^q$, so $n^q a_n$ is eventually non-increasing and $a_n \leqslant Cn^{-q}$. If $p < 1$, eventually $\frac{a_{n+1}}{a_n} \geqslant \frac{n}{n+1}$, so na_n is eventually non-decreasing and $a_n \geqslant c/n$.

Hint 2. The case $p = 1$ (Gauss): $\ln \frac{(n+1)a_{n+1}}{na_n} = \ln\left(1 + \frac{1}{n}\right) + \ln\left(1 - \frac{1}{n} + O(n^{-2})\right) = O(n^{-2})$ is summable, so $na_n \to L > 0$ and the series diverges like $\sum \frac{1}{n}$.

Hint 3. Here $\frac{a_{n+1}}{a_n} = \left(\frac{2n+1}{2n+2}\right)^p = \left(1 - \frac{1}{2n+2}\right)^p = 1 - \frac{p}{2n} + O(n^{-2})$, which tends to 1 for every p. So the series converges for $p > 2$ and diverges for $p \leqslant 2$ (for $p = 2$ by the Gauss case). The same answer follows from Wallis: $\frac{(2n-1)!!}{(2n)!!} \sim \frac{1}{\sqrt{\pi n}}$.

Problem 2

Hint 1. Both convergent series come from AM–GM: $\sqrt{a_n a_{n+1}} \leqslant \frac{a_n + a_{n+1}}{2}$ and $\frac{\sqrt{a_n}}{n} \leqslant \frac{1}{2}\left(a_n + \frac{1}{n^2}\right)$. For the example take $a_n = \frac{1}{n^2}$.

Hint 2. For the last part one example works for all $p \leqslant \frac{1}{2}$. You need a_n that is summable, but only barely, so that $\sqrt{a_n}$ is about $\frac{1}{\sqrt{n}}$.

Hint 3. $a_n = \frac{1}{n \ln^2 n}$ $(n \geqslant 2)$ gives $\frac{\sqrt{a_n}}{n^p} = \frac{1}{n^{p+1/2} \ln n} \geqslant \frac{1}{n \ln n}$ for $p \leqslant \frac{1}{2}$, and $\sum \frac{1}{n \ln n}$ diverges. (For $p > \frac{1}{2}$ Cauchy–Schwarz with $\sum n^{-2p}$ gives convergence, so $\frac{1}{2}$ is the threshold.)

Problem 3

Hint 1. Put $s = \frac{1}{1-\alpha} > 1$ and split $x_k^s = x_k \cdot x_k^{s-1} = x_k \cdot x_k^{\alpha/(1-\alpha)}$. It is enough to show that $x_k^{\alpha/(1-\alpha)} \leqslant Ck^{-\alpha}$, i.e. that $x_k \leqslant C' k^{-(1-\alpha)}$.

Hint 2. This bound uses the monotonicity: $\sum_{k \leqslant n} \frac{x_k}{k^\alpha} \geqslant x_n \sum_{k \leqslant n} k^{-\alpha} \geqslant c\, x_n n^{1-\alpha}$, and the left-hand side is bounded.

Hint 3. Hence $x_k^s \leqslant C\, \frac{x_k}{k^\alpha}$, and the series converges by comparison. (Applying the same estimate to the block $\frac{n}{2} < k \leqslant n$ even gives $x_n n^{1-\alpha} \to 0$, but boundedness is all you need.)

Problem 4

Hint 1. Look for a telescoping bound $\frac{1}{\sqrt{n}\,(n+1)} < c\left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right)$. Which c gives the sum 2?

Hint 2. $2\left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}\right) = \frac{2}{\sqrt{n}\sqrt{n+1}\,(\sqrt{n} + \sqrt{n+1})}$.

Hint 3. Comparing with $\frac{1}{\sqrt{n}\,(n+1)}$ reduces to $\sqrt{n} + \sqrt{n+1} < 2\sqrt{n+1}$, which is true. Summing over $n \geqslant 1$ gives $\sum < 2 \cdot \frac{1}{\sqrt{1}} = 2$.

Problem 5

Hint 1. Find the growth of $I_n = \int_0^n |\sin \pi x|^x \, dx$. On $[k, k+1]$ the exponent lies between k and $k+1$, and $|\sin \pi x| \leqslant 1$, so

$$J_{k+1} \leqslant \int_k^{k+1} |\sin \pi x|^x \, dx \leqslant J_k, \qquad J_k = \int_0^1 \sin^k(\pi t) \, dt.$$

Hint 2. $J_k = \frac{2}{\pi} \int_0^{\pi/2} \sin^k \theta \, d\theta \sim \frac{2}{\pi} \sqrt{\frac{\pi}{2k}} = \sqrt{\frac{2}{\pi k}}$ (Wallis integrals). For the answer only the order $J_k \asymp k^{-1/2}$ is needed.

Hint 3. Hence $I_n \sim \sum_{k<n} \sqrt{\frac{2}{\pi k}} \sim 2\sqrt{\frac{2}{\pi}}\, \sqrt{n}$, and $a_n \sim 2\sqrt{\frac{2}{\pi}}\, n^{1/2-p}$. The series converges exactly for $p > \frac{3}{2}$.

Problem 6

Hint 1. Get rid of the root: $2x_{n+1} - x_n - 1 = \sqrt{(x_n+1)^2 + 4}$. Square it and simplify.

Hint 2. You get $x_{n+1}^2 - x_{n+1}x_n - x_{n+1} - 1 = 0$. Also $x_{n+1} \geqslant x_n + 1$, so $x_n \to \infty$. Try to write $\frac{1}{x_{n+1}^2 - 1}$ as a difference $g(x_n) - g(x_{n+1})$.

Hint 3. The relation says $x_{n+1}^2 - 1 = x_{n+1}(x_n + 1)$ and $x_{n+1}(x_{n+1} - x_n) = x_{n+1} + 1$. Together they give

$$\frac{1}{x_n + 1} - \frac{1}{x_{n+1} + 1} = \frac{x_{n+1} - x_n}{(x_n+1)(x_{n+1}+1)} = \frac{1}{x_{n+1}(x_n+1)} = \frac{1}{x_{n+1}^2 - 1}.$$

Hint 4. So $y_n = \frac{1}{x_1^2 - 1} + \frac{1}{x_1 + 1} - \frac{1}{x_n + 1} \to \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

Problem 7

Hint 1. Put $r_n = \frac{x_{n+1}}{x_n} \geqslant 1$. The term is $\sqrt{r_n} - \frac{1}{\sqrt{r_n}} = \frac{r_n - 1}{\sqrt{r_n}} \geqslant 0$.

Hint 2. It converges: $\frac{r_n - 1}{\sqrt{r_n}} \leqslant r_n - 1 = \frac{x_{n+1} - x_n}{x_n} \leqslant \frac{x_{n+1} - x_n}{x_1}$, and these bounds telescope to at most $\frac{\lim x_n - x_1}{x_1}$.

Problem 8

Hint 1. F is increasing, $F(1) = 0$, $F(x) \to \infty$, so start the sums at $n = 2$. Monotonicity of f gives $\int_n^{n+1} f \leqslant f(n) \leqslant \int_{n-1}^n f$, i.e. $F(n+1) - F(n) \leqslant f(n) \leqslant F(n) - F(n-1)$.

Hint 2. Divergence: $\frac{f(n)}{F(n)} \geqslant \frac{F(n+1) - F(n)}{F(n)} \geqslant \ln \frac{F(n+1)}{F(n)}$ (use $u - 1 \geqslant \ln u$), and the logarithms telescope to $\ln F(N) \to \infty$.

Hint 3. Convergence: for $t \in [n-1, n]$ we have $f(n) \leqslant f(t)$ and $F(n) \geqslant F(t)$, so $\frac{f(n)}{F(n)^{1+\varepsilon}} \leqslant \int_{n-1}^n \frac{f(t)}{F(t)^{1+\varepsilon}} \, dt$. Summing over $n \geqslant 3$ gives at most $\int_2^\infty \frac{F'}{F^{1+\varepsilon}} = \frac{F(2)^{-\varepsilon}}{\varepsilon}$.

Problem 9

Hint 1. In all three cases start from $n^{1/n} = e^{\ln n / n} \to 1$, and $n^{1/n} - 1 \sim \frac{\ln n}{n}$.

Hint 2. $\frac{1}{n^{1+1/n}} = \frac{1}{n} \cdot \frac{1}{n^{1/n}} \sim \frac{1}{n}$: limit comparison with the harmonic series, divergent. $(n^{1/n} - 1)^p \sim \frac{\ln^p n}{n^p}$: limit comparison with the scale, convergent exactly for $p > 1$.

Hint 3. For $\left(\sqrt[n]{n}-1\right)^n$ the root test is the cheapest: the n-th root of the term is $\sqrt[n]{n}-1\to 0$, so the series converges.

Problem 10

Hint 1. Put $r_n=|A_0A_n|$. The triangle $A_0A_{n-1}A_n$ has its right angle at A_{n-1}, so $r_n^2=r_{n-1}^2+a_n^2$, i.e. $r_n^2=a_1^2+\cdots+a_n^2$. The angle is $\theta_n=\angle A_{n-1}A_0A_n=\arctan\frac{a_n}{r_{n-1}}$.

Hint 2. The r_n increase, so $\{A_n\}$ is unbounded iff $r_n\to\infty$, i.e. iff $\sum a_n^2=\infty$. If $\sum\theta_n$ converges, then $u_n=\frac{a_n}{r_{n-1}}\to 0$ and $\theta_n\sim u_n$, so $\sum u_n$ converges, and so does $\sum u_n^2$.

Hint 3. But $u_n^2=\frac{r_n^2-r_{n-1}^2}{r_{n-1}^2}\geqslant\frac{r_n^2-r_{n-1}^2}{r_n^2}$, and for $r_n^2\to\infty$ this series diverges (Abel–Dini: the block $n<k\leqslant m$ adds at least $\frac{r_m^2-r_n^2}{r_m^2}$, which tends to 1 as $m\to\infty$). So it is not possible.